MIL–F–50086A
29 DECEMBER 1960
SUPERSEDING
MIL–F–50086 (CmlC)
10 SEPTEMBER 1959

MILITARY SPECIFICATION

FILTER UNIT, GAS-PARTICULATE, TANK, THREE-MAN, 12-CFM, M8A2

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a gas-particulate filter unit and installation kit.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

UU–T–116 — Tape, Paper, Gummed, Water-Resistant.
PPP–B–636 — Boxes, Fiber.
PPP–T–60 — Tape; Pressure-Sensitive Adhesive, Waterproof for Packaging and Sealing.
PPP–T–76 — Tape, Pressure-Sensitive Adhesive, Paper, Water-Resistant

MILITARY

MIL–P–116 — Preservation, Methods of.
MIL–B–121 — Barrier-Material, Greaseproofed, Flexible, (Waterproofed).
MIL–B–130 — Barrier-Material, Paper, Noncorrosive.
MIL–P–50088 — Purifier, Air, M2A1.
MIL–I–50089 — Installation Kit, Gas, Gas-Particulate Filter Unit, Armored Vehicle, M20

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

CHEMICAL CORPS

C5–19–1501 — Filter Unit, Gas-Particulate, Tank, 3-Man, 12 CFM, M8A2—Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the ex-

571048-61—(776)

FSC 4240

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

tent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components. All materials and components used in the manufacture of this filter unit shall conform to the specifications and drawings cited on Drawing C5-19-1501.

3.2 Wormanship. Poor workmanship shall not be allowed. Damage such as bent, burred, or chipped metal sections and contamination by foreign matter inside or outside shall be indicative of poor workmanship.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Objective evidence. The supplier shall present objective evidence as required by the Government representative that all requirements of section 3 and of section 5 have been met (see 6.5).

4.3 Alternative inspection (including testing) procedures. The supplier may utilize any alternative inspection procedure which will assure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to its institution. In case of dispute, the procedures of this specification will govern.

4.4 Inspection provisions.

4.4.1 *Lotting.* A lot shall consist of the filter units offered for procurement at one time by one manufacturer. No lot shall contain more than one lot of air purifiers.

4.4.2 *Sampling.* Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.4.3 Examination. Sample filter units, level A and level B preparation for delivery requirements thereof shall be examined in accordance with the classification of defects and with Standard MIL-STD-105.

4.4.4 *Classification of defects.*

4.4.4.1 *Filter unit, gas-particulate, tank, 3-man, M8A2.* (Drawing C5-19-1501, Bill of Materials.)

Categories *Defects*

Critical: None defined.

Major:

AQL 1.0 percent defective.

101. Component missing, incorrect, or incorrectly assembled.
102. Marking incorrect or missing.
103. Workmanship.

4.4.4.2 *Preparation for delivery* (sec. 5)

Categories *Defects*

Critical: None defined.

Major:

AQL 2.5 percent defective.

101. Packaging methods incorrect.
102. Containers incorrect.
103. Components or technical publications missing or not packaged in containers as specified.
104. Internal cushioning or blocking missing or insufficient to make a tight package.
105. Containers closure incorrect.
106. Containers damaged (broken, torn, or punctured).
107. Strapping loose or missing, where specified.
108. Marking incorrect, missing, or illegible.

Minor:

AQL 6.5 percent defective

201. Contamination.
202. Preservatives missing, where specified.
203. Hollow circular fiberboard core missing, where specified.
204. Attached hose fittings not cushion-wrapped as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The filter unit and components thereto shall be cleaned and dried in accordance with Specification MIL–P–116, using cleaning method C–1 and any of the drying methods listed therein.

5.2 Preservation and packaging.

5.2.1 *Level A.*

5.2.1.1 *Unit preservation and packaging.*

5.2.1.1.1 *Hose, 6- and 9-foot lengths.* Fittings attached to hose shall be cushion-wrapped with either type II, grade A, class 2, greaseproof barrier material or type II, noncorrosive barrier material conforming to Specification MIL–B–121 or MIL–B–130 respectively. The wraps shall be secured in place by either tying or taping.

5.2.1.1.2 *Spring clamps and hose loop clamps.* The spring clamps and hose loop clamps shall be packaged together in accordance with method IC–3 of Specification MIL–P–116. (See 5.2.1.2.1 for quantity per package.)

5.2.1.1.3 *Cable loop clamps, screw, lockwashers, and flat washers.* The cable loop clamps, screws, lockwashers, and flat washers shall be preserved with type P–7 preservative oil of Specification MIL–P–116, and packaged together in accordance with method IC–1 of the same specification (see 5.2.1.2.1 for quantity per package).

5.2.1.1.4 *Installation kit.* The kit shall be preserved and unit packaged as specified in Specification MIL–I–50089.

5.2.1.1.5 *Air purifier.* The air purifier shall be packaged as specified in Specification MIL–P–50088.

5.2.1.1.6 *Technical publications.* The technical publications shall be packaged in accordance with method IC–3 of Specification MIL–P–116.

5.2.1.2 *Intermediate packaging.*

5.2.1.2.1 *Container I.* The contents of container I shall consist of: two 6-foot lengths of hose, one 9-foot length of hose, three spring clamps, 10 hose loop clamps, fine cable loop clamps, 20 screws, 17 lockwashers, 12 flat washers, and technical publications. These items preserved and unit packaged as specified in 5.2.1.1.1, 5.2.1.1.2, 5.2.1.1.3, and 5.2.1.1.6 shall be intermediately packaged together in a W5c, style RSC, fiberboard box, conforming to Specification PPP–B–636. The required lengths of hose shall be loosely coiled within the box around a circular fiberboard core, that has been fabricated from the same material as the box. The unit packaged clamps, screws, and washers shall then be placed inside of the fiberboard core. Internal cushioning and blocking shall be used as required to make a tight package. The packaged technical publications shall then be placed inside of the box on top of the packed items. The top and bottom flaps of the fiberboard box shall be closed by taping in accordance with the appendix to Specification PPP–B–636 for interior boxes using gummed or pressure-sensitive tape conforming to Specification PPP–T–76 or UU–T–116.

5.2.1.2.2 *Container II.* The contents of container II shall consist of one installation kit intermediately packaged as specified in Specification MIL–I–50089.

5.2.1.2.3 *Container III.* The contents of container III shall consist of one air purifier packaged as specified in 5.2.1.1.5.

5.3 Packing. Levels shall be as specified (see 6.2).

5.3.1 *Level A.* The filter unit, consisting of items preserved and packaged as specified in 5.2, shall be packed in a V3c, style RSC, fiberboard box conforming to Specification PPP–B–636. Containers II and III shall be placed alongside each other in the bottom of the box with container I placed on top of these containers. The box shall be closed and strapped in accordance with the appendix requirements of Specification PPP–B–636 for exterior containers. Galvanized strapping shall be used. In addition, all box seams, including the manufacturer's joint, shall be taped with minimum 3-inch wide type III, class 1, color optional, pressure-sensitive tape conforming to

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.

Specification PPP–T–60, extending the tape over all corners and edges at least 3 inches.

5.3.2 *Level B.* The filter unit shall be placed as specified in 5.3.1 with the exception that coated-finish strapping shall be used.

5.3.3 *Level C.* The filter unit shall be packed as specified in 5.3.1, except that the shipping container shall conform to a type II, class 1, fiberboard box of Specification PPP–B–636, closing the box in accordance with the appendix requirements thereto for domestic boxes.

5.4 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Intended use. This filter unit is designed to remove toxic aerosols and gas from air supplied to tank crews.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

6.3 Stock number. The Federal stock number for this item is 4240–691–1505.

6.4 Component specifications. The component specifications for the M8A2 Filter Unit are as follows:

Title	*Number*
Filter, gas, M12A1	MIL–F–14512B
Filter, particulate, 12cFM, M13	MIL–F–52011B
Hose: for filter units	MIL–H–14506B
Installation kit, M20	MIL–I–50089A
Motor blower	MIL–M–50084A
Precleaner, air purifier, M1A1	MIL–P–50085A
Purifier, air, M2A1	MIL–P–50088A
Separator, dust	MIL–S–50087A

6.5 Objective evidence. Provisions for objective evidence and inspection records, and maintenance of inspection records will be specified by the procuring activity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, on in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:

Army—Cm1C
Navy—*MC*

Preparing activity:

Army—Cm1C

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:16Z
Check the source to verify that this is the current version before use.